RRVM — Rollback & Recovery Module

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OOF® Origin Open Foundation®

Independent Methodological Authority

Architecture: SLGA™ — Software Lifecycle Governance Architecture

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Subcategory: Software Deployment Governance

Type: Core Module

Governed Space: Software Deployment Rollback & Recovery Governance

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Canonical Definition

RRVM — Rollback & Recovery Module is the Core Module within SDPS that governs how software deployment failures, unacceptable operational conditions, and material deployment-state deviations are addressed through legitimate, feasible, controlled, traceable, and verifiable rollback or recovery actions.

RRVM establishes rollback and recovery as distinct but interconnected

governance mechanisms for restoring or establishing an acceptable operational state when deployment conditions cannot be maintained.

It governs the identification of corrective-action triggers, the assessment

of rollback and recovery feasibility, the selection and authorization of corrective actions, their execution boundaries, the verification of resulting operational conditions, and the treatment of unresolved consequences.

The module distinguishes restoration of an earlier software artifact from

restoration of an acceptable operational state, execution of corrective procedures from verified recovery, and technical recovery capability from legitimate authority to intervene.

RRVM recognizes that some deployment consequences are only partially reversible, that restoring previous software may introduce new incompatibilities, and that recovery may require forward correction rather than reversal.

It applies to human-operated deployments, automated delivery pipelines,

autonomous AI agents, distributed infrastructure, cloud services, industrial systems, embedded software, and robotic platforms.

Its principal governance output is the Software Deployment Recovery Profile.

Operational Role

RRVM establishes the governance conditions under which deployment-related

operational failures and unacceptable states may be corrected.

It governs:

• Rollback and recovery case identification

• Deployment failure association

• Corrective-action trigger assessment

• Current operational state identification

• Previous and target state references

• Rollback feasibility

• Recovery feasibility

• Reversibility and irreversible consequences

• Corrective strategy selection

• Corrective-action authority

• Execution actor eligibility

• Automated and autonomous intervention boundaries

• Recovery prerequisites

• Corrective execution constraints

• Service and data continuity

• Rollback and recovery verification

• Residual risk and unresolved conditions

• Corrective-action failure

• Escalation and alternative recovery

• Recovery-state classification

• Historical evidence and continuity

• Handoff to deployment monitoring and lifecycle governance

RRVM does not independently authorize the original software deployment, redefine software release validity, or treat restoration of a previous version as automatic proof of operational recovery.

Operational Question

How can software deployment failures or unacceptable operational conditions be addressed through legitimate, feasible, traceable, and verifiable rollback or recovery actions while preserving operational constraints, historical states, residual risks, and continuity of software lifecycle governance?

Position in Architecture Governance Flow

SLGA™ Governance Flow

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-cutting: Software Lifecycle Control

SDPS Core Module Flow

DPM — Deployment Planning Module → DAM — Deployment Authorization Module →

PVCM — Production Version Control Module → DSMM — Deployment State Monitoring Module → RRVM — Rollback & Recovery Module

RRVM is the fifth Core Module within SDPS.

It receives recovery planning references from DPM, applicable authority

references from DAM, production version identities from PVCM, and deployment-state observations and intervention triggers from DSMM.

It governs the selection, authorization applicability, execution

conditions, and verification of corrective deployment actions.

Its outputs return the resulting version and operational state evidence to

PVCM and DSMM and preserve relevant references for subsequent software change management and lifecycle control.

• Governed Space

Software Deployment Rollback & Recovery Governance

This governed space concerns the legitimate and evidence-supported

correction of unacceptable operational states associated with software deployment.

It begins when a deployment-related condition creates a justified requirement to assess rollback, recovery, or another corrective response.

It continues through state assessment, strategy selection, authority verification, feasibility evaluation, corrective execution, operational verification, residual-condition treatment, and closure.

The governed space includes corrective actions initiated during deployment,

immediately after deployment, or later when an unacceptable state can be materially associated with a deployment transition.

It does not replace general incident management, disaster recovery, business continuity, or software change governance outside the applicable deployment-related scope.

Governed Object

The primary Governed Object is:

A Software Deployment Recovery Case representing the evidence-supported

relationship between an identifiable deployment, an unacceptable or insufficiently established operational state, a justified corrective objective, an applicable rollback or recovery strategy, legitimate execution authority, corrective-action evidence, and the resulting verified or unresolved operational conditions.

The governed object must preserve:

Recovery Case ID

Deployment ID

Environment ID

Trigger Reference

Current Operational State

Previous State Reference

Recovery Objective

Selected Corrective Strategy

Authority Reference

Execution Actor Reference

Recovery Prerequisites

Corrective Execution Record

Resulting Production Version Reference

Recovery Verification

Residual Conditions

Recovery Classification

Version

Scope

RRVM governs:

Recovery Case Formation

Corrective Trigger Assessment

Operational State Identification

Recovery Objective Definition

Rollback Suitability

Recovery Strategy Selection

Feasibility Assessment

Corrective-Action Authorization

Execution Scope and Boundaries

Recovery Prerequisites

Data and Configuration Restoration

Dependency Compatibility

Service Continuity

Corrective Execution Evidence

Recovery Verification

Residual Risk

Recovery Failure

Escalation

Recovery Closure

Historical Continuity

RRVM does not prescribe a particular backup system, deployment platform, infrastructure provider, recovery tool, or incident response methodology.

Foundational Principle

Rollback and recovery are governed by the operational state they can legitimately and verifiably establish—not merely by the technical actions they can execute.

Rollback Executed ≠ Operational Recovery Verified.

Previous Software Version Restored ≠ Previous Operational State Restored.

Recovery Procedure Available ≠ Recovery Feasible.

Technical Intervention Capability ≠ Legitimate Intervention Authority.

Service Restarted ≠ Service Recovered.

Recovery Case Identity

Every governed rollback or recovery case must have a stable identity.

Relevant elements include:

Recovery Case ID

Deployment ID

Environment ID

Affected Software Product

Trigger ID

Case Creation Time

Responsible Recovery Function

Current Case State

Evidence Reference

Case Version

A recovery case may involve multiple corrective attempts without losing its historical identity.

Recovery Trigger

A recovery trigger is an observed condition requiring assessment of corrective action.

Examples include:

Deployment Execution Failure

Operational Acceptance Failure

Critical Service Degradation

Production Version Mismatch

Configuration Failure

Dependency Incompatibility

Data Integrity Failure

Security-Relevant Deployment Condition

Unauthorized Deployment Action

Unacceptable Physical-System Behavior

Recovery Trigger Detected ≠ Rollback Automatically Authorized.

Trigger Evidence

RRVM must preserve the evidence supporting corrective-action consideration.

Relevant elements include:

Trigger ID

Observed Condition

Observation Source

Observation Time

Affected Environment

Affected Operational Segments

Severity

Evidence Confidence

Applicable Threshold

Known Uncertainty

The trigger must be sufficiently characterized to support the required corrective assessment.

Deployment Association

RRVM must identify the deployment to which the recovery case relates.

Relevant references include:

Deployment ID

Release ID

Deployment Plan ID

Deployment Authorization ID

Production Version Reference

Deployment State Reference

Execution History

Material Deviations

A condition occurring after deployment must not automatically be attributed to that deployment without adequate evidence.

Current Operational State

Before selecting corrective action, RRVM must establish the current operational state to the extent necessary for safe and effective intervention.

Relevant elements include:

Current Production Version

Current Configuration

Current Dependency State

Current Data State

Service Availability

Infrastructure Condition

Security Condition

Operational Restrictions

Known Failures

Unverified Conditions

An incomplete state assessment must remain explicit.

Previous Operational State

RRVM must identify relevant historical operational states when rollback is considered.

Relevant elements include:

Previous Software Version

Previous Artifact Identity

Previous Configuration

Previous Dependency State

Previous Data State

Previous Infrastructure State

Previous Service Conditions

Historical Evidence

A historical state may be documented without remaining technically reproducible.

Recovery Objective

Every corrective action must have an identifiable recovery objective.

Potential objectives include:

Restore Service Availability

Restore an Authorized Production Version

Restore Compatible Configuration

Restore Data Integrity

Restore Dependency Compatibility

Remove an Unauthorized Artifact

Contain Operational Degradation

Establish a Safe Restricted State

Complete a Forward Correction

Restore Verified Operational Acceptance

The objective must identify the required operational result rather than merely name the intended corrective procedure.

Rollback Definition

Rollback is a corrective deployment action intended to reverse specified changes or return selected software, configuration, infrastructure, or operational elements toward an earlier reference state.

Rollback may include:

Software Artifact Reversion

Configuration Reversion

Traffic Reversion

Infrastructure Reversion

Feature Activation Reversal

Data Restoration Where Feasible

Rollback may be partial and does not necessarily restore the entire previous operational state.

Recovery Definition

Recovery is the governed process of establishing an acceptable operational state following a deployment-related failure or unacceptable condition.

Recovery may involve:

Rollback

Forward Correction

Configuration Repair

Dependency Restoration

Infrastructure Repair

Data Restoration

Service Reconfiguration

Operational Containment

Alternative Service Activation

Controlled Degraded Operation

Recovery is defined by the resulting operational condition, not by whether an earlier software version is restored.

Rollback–Recovery Distinction

Rollback and recovery must remain separate governance concepts.

Rollback concerns reversal of specified deployment-related changes.

Recovery concerns establishment of an acceptable operational state.

A rollback may be one component of recovery.

A recovery may succeed without rollback.

A rollback may execute successfully while recovery fails.

Rollback Success ≠ Recovery Success.

Corrective Strategy Selection

RRVM must evaluate the suitability of available corrective strategies.

Potential strategies include:

Full Rollback

Partial Rollback

Forward Fix

Configuration Restoration

Dependency Correction

Data Restoration

Traffic Redirection

Service Isolation

Restricted Operation

Infrastructure Replacement

Emergency Recovery Deployment

Strategy selection must reflect the current operational reality and the intended recovery objective.

Strategy Suitability

Corrective strategy suitability depends on:

Failure Characteristics

Operational Consequences

Current State

Reversibility

Dependency Compatibility

Data Compatibility

Resource Availability

Execution Risk

Recovery Time Constraints

Verification Capability

Residual Risk

The technically fastest response is not necessarily the most appropriate governed response.

Rollback Feasibility

Rollback feasibility evaluates whether specified changes can be reversed under actual operational conditions.

Relevant elements include:

Previous Artifact Availability

Configuration Compatibility

Dependency Compatibility

Data Compatibility

Infrastructure Availability

Rollback Tool Capability

Required Access

Operational Timing

Expected Service Impact

Verification Capability

A documented rollback procedure does not establish current feasibility.

Recovery Feasibility

Recovery feasibility evaluates whether the selected corrective strategy can reasonably establish the required operational state.

Relevant elements include:

Recovery Resources

Required Expertise

Infrastructure Capability

Data Availability

Dependency Conditions

Security Restrictions

Operational Constraints

Recovery Time

Alternative Strategies

Verification Capability

Recovery feasibility must be reassessed when material conditions change.

Reversibility Assessment

RRVM must identify which deployment consequences are reversible, partially reversible, or irreversible.

Relevant categories include:

Software Artifact Changes

Configuration Changes

Database Schema Changes

Persistent Data Mutations

External System Actions

Credential Changes

Infrastructure Modifications

Firmware Updates

Physical-System State Changes

Reversibility must not be inferred from the existence of a previous software version.

Irreversible Consequences

Some deployment consequences cannot be fully reversed.

Examples include:

Deleted Data Without Valid Backup

Destructive Schema Transformation

External Financial Transaction

Irreversible Firmware Modification

Permanent Credential Invalidation

Physical Action Already Performed

Externally Published Information

RRVM must identify the remaining consequences and establish appropriate containment, compensation, forward correction, or alternative recovery references where applicable.

Recovery Prerequisites

RRVM must identify conditions required before corrective execution.

These may include:

Recovery Objective Established

Current State Sufficiently Identified

Corrective Strategy Selected

Authority Applicable

Execution Actor Eligible

Required Artifacts Available

Backup Integrity Established

Dependency Compatibility Assessed

Recovery Resources Available

Verification Capability Available

Mandatory prerequisites must not be silently bypassed.

Prerequisite Condition States

RRVM applies the following condition model:

Satisfied

Not Satisfied

Partially Satisfied

Unknown

Not Applicable

Exception

Each condition must retain supporting evidence or an applicable exception reference.

Unknown ≠ Satisfied.

Backup Exists ≠ Backup Usable.

Exception Recorded ≠ Exception Authorized.

Corrective-Action Authority

RRVM requires legitimate authority for rollback or recovery actions.

Relevant elements include:

Authority Source

Recovery Decision Actor

Execution Actor

Permitted Corrective Actions

Target Environment

Operational Scope

Delegation Conditions

Emergency Conditions

Validity

Restrictions

Original deployment authorization does not automatically establish unrestricted authority for corrective actions.

Recovery Decision Authority

RRVM must distinguish authority to select a recovery strategy from authority to execute it.

Relevant decision functions include:

Approve Rollback

Approve Forward Recovery

Authorize Data Restoration

Authorize Service Shutdown

Authorize Infrastructure Replacement

Authorize Restricted Operation

Approve Recovery Closure

Decision authority must remain applicable to the operational consequences of the proposed action.

Execution Actor Eligibility

Corrective actions may be executed by:

Human Operators

Recovery Teams

Automated Pipelines

Infrastructure Orchestrators

Autonomous AI Agents

External Service Providers

RRVM must identify the actor's applicable permission, capability, operational scope, and execution restrictions.

Recovery Capability ≠ Recovery Authority.

Automated Rollback

Automated rollback may be initiated when predefined conditions are satisfied.

RRVM must preserve:

Automation Identity

Trigger Conditions

Authorized Rollback Scope

Rollback Target

Execution Preconditions

Execution Limits

Stop Conditions

Verification Requirements

Escalation Conditions

Execution Evidence

Automated rollback must not bypass mandatory authority or operational feasibility requirements.

Autonomous Recovery Agents

Autonomous agents may recommend, plan, or execute corrective actions within legitimate boundaries.

Relevant elements include:

Agent Identity

Assigned Recovery Function

Authority Reference

Permitted Tool Paths

Permitted Infrastructure Actions

Data Access Boundaries

Recovery Constraints

Human Override

Escalation Conditions

Evidence Requirements

An agent's objective to restore service does not authorize every action that could contribute to that objective.

Constituent Recovery Action Authority

A recovery strategy may require multiple distinct actions.

Examples include:

Retrieve Previous Artifact

Modify Deployment Configuration

Restore Database Backup

Change Network Routing

Disable a Service

Replace Infrastructure

Rotate Credentials

Activate Alternative Environment

Restart Physical Equipment

Each material action must remain within applicable authority.

Recovery Objective Authority ≠ Authority for Every Recovery Action.

Emergency Recovery

Emergency conditions may justify accelerated corrective decisions.

RRVM must preserve:

Emergency Trigger

Emergency Authority

Urgency Justification

Permitted Corrective Scope

Mandatory Safety Conditions

Temporary Exceptions

Decision Evidence

Validity

Post-Recovery Review

Urgency does not independently create unlimited intervention authority.

Recovery Execution Boundary

The recovery execution boundary defines the permitted extent of corrective

actions.

Relevant dimensions include:

Affected Software

Target Environment

Operational Segments

Infrastructure Resources

Data Operations

Configuration Changes

External System Interactions

Execution Tools

Time

Service Impact

Material actions outside the established boundary require applicable

reassessment or authorization.

Corrective Execution Sequence

RRVM must establish the execution sequence where order materially affects recovery.

Potential stages include:

Containment

State Preservation

Backup Verification

Service Isolation

Artifact Restoration

Configuration Restoration

Dependency Restoration

Data Restoration

Service Activation

Operational Verification

Recovery Closure

The sequence must reflect dependencies and operational constraints.

Recovery Checkpoints

Recovery checkpoints establish conditions that must be evaluated before corrective execution continues.

Relevant elements include:

Checkpoint ID

Execution Stage

Required Condition

Evidence

Decision Authority

Permitted Continuation

Failure Response

Escalation Requirement

Checkpoints are especially important where corrective actions may be destructive or difficult to reverse.

Recovery Stop Conditions

RRVM must identify conditions requiring corrective execution to stop or pause.

Examples include:

Authority No Longer Applicable

Unexpected Data State

Artifact Identity Mismatch

Backup Verification Failure

Dependency Incompatibility

Unacceptable Service Impact

Uncontrolled Infrastructure Change

Loss of Required Monitoring

Recovery Action Outside Permitted Scope

A recovery procedure must not continue solely because its automation remains executable.

Service Continuity During Recovery

Recovery actions may create additional service disruption.

RRVM must evaluate:

Permitted Downtime

Critical Service Requirements

Traffic Management

Redundancy

Failover

Temporary Degradation

Affected Users

External Dependencies

Operational Safety

Alternative Service Availability

Corrective action must account for the operational consequences it introduces.

Data Recovery Governance

Where recovery affects persistent data, RRVM must identify:

Data State Before Deployment

Current Data State

Backup Reference

Backup Integrity

Restoration Point

Data Loss Exposure

Schema Compatibility

External Data Dependencies

Restoration Verification

Residual Data Conditions

A database backup must not be assumed usable without appropriate evidence.

Configuration Recovery

Configuration recovery must preserve the relationship between intended and restored configuration states.

Relevant elements include:

Previous Configuration Reference

Current Configuration

Target Recovery Configuration

Configuration Dependencies

Application Sequence

Verification Method

Known Differences

Residual Restrictions

Configuration restoration may require a state different from the historical baseline when other operational elements have changed.

Dependency Recovery

RRVM must identify dependency conditions affecting recovery.

Relevant elements include:

Required Dependency Versions

Available Dependency Versions

External Service Compatibility

Infrastructure Dependencies

Network Dependencies

Runtime Compatibility

Recovery Sequence

Verification Evidence

Restoring a previous application version may fail when required dependencies have changed or become unavailable.

Infrastructure Recovery

Recovery may require infrastructure restoration or replacement.

Relevant elements include:

Affected Infrastructure

Previous Infrastructure Reference

Target Recovery Infrastructure

Capacity Requirements

Network Conditions

Security Configuration

Resource Availability

Deployment Compatibility

Operational Verification

Infrastructure recovery must preserve the applicable operational and authorization boundaries.

Production Version Restoration

RRVM must preserve the relationship between the intended recovery version and the version actually established.

Relevant elements include:

Previous Production Version

Recovery Target Version

Recovery Artifact Identity

Restored Artifact Identity

Environment

Operational Segment

Activation State

Verification Evidence

PVCM governs the resulting production version identity and consistency.

Recovery State Monitoring

RRVM must obtain evidence about corrective execution and its operational consequences.

Relevant observations include:

Corrective Execution Progress

Artifact Restoration

Configuration State

Dependency State

Data Integrity

Service Availability

Operational Health

Security Conditions

Residual Failures

Monitoring Coverage

DSMM governs the resulting deployment-state observation and classification.

Corrective Execution Evidence

RRVM must preserve evidence of actions actually performed.

Relevant elements include:

Recovery Action ID

Execution Actor

Authority Reference

Target Resource

Executed Operation

Execution Time

Input State Reference

Resulting State Reference

Execution Result

Evidence Source

A successful tool response does not independently prove that the required operational result was achieved.

Recovery Verification

Recovery verification evaluates whether the corrective objective has been

achieved.

Relevant dimensions include:

Production Version Identity

Configuration Compatibility

Dependency Compatibility

Data Integrity

Service Availability

Required Functionality

Security Conditions

Operational Restrictions

Monitoring Coverage

Evidence Sufficiency

Recovery verification must address the resulting operational state rather than only the completion of corrective procedures.

Recovery Acceptance Criteria

RRVM must identify the conditions required for recovery acceptance.

These may include:

Required Services Restored

Authorized Production Version Established

Required Configuration Verified

Data Integrity Supported

Dependencies Compatible

Critical Errors Resolved

Operational Restrictions Satisfied

Monitoring Sufficient

Residual Risks Evaluated

Recovery Evidence Complete

Acceptance criteria must reflect the recovery objective and applicable operational conditions.

Partial Recovery

A partial recovery exists when some required operational conditions are restored while others remain unresolved.

Relevant elements include:

Restored Functions

Unrestored Functions

Affected Operational Segments

Residual Failures

Temporary Restrictions

Additional Recovery Requirements

Evidence

Partial recovery must not be classified as complete solely because the most visible service has resumed.

Restricted Recovery State

An operational environment may be restored to a restricted but acceptable condition.

Examples include:

Read-Only Service

Limited User Access

Reduced Capacity

Disabled Noncritical Functions

Isolated Infrastructure Segment

Manual Operating Mode

Restricted Robotic Operation

Such a state must preserve its restrictions, validity, monitoring requirements, and unresolved recovery obligations.

Recovery Failure

Recovery failure occurs when the selected corrective strategy cannot establish the required operational state.

Examples include:

Rollback Artifact Unavailable

Data Restoration Failure

Dependency Incompatibility

Configuration Restoration Failure

Infrastructure Recovery Failure

Verification Failure

New Operational Degradation

Unacceptable Residual Condition

Recovery failure must trigger reassessment rather than an unsupported declaration of success.

Secondary Recovery

Where an initial corrective strategy fails, RRVM may govern a subsequent recovery strategy.

Relevant elements include:

Failed Strategy Reference

Failure Evidence

Current Operational State

Revised Recovery Objective

Alternative Strategy

New Prerequisites

Authority Applicability

Revised Execution Boundaries

Verification Requirements

The secondary strategy must be evaluated against the actual state resulting from earlier corrective actions.

Recovery-Induced Change

Corrective actions may introduce new software or operational changes.

Examples include:

Forward Fix Deployment

New Configuration Baseline

Dependency Replacement

Infrastructure Reconfiguration

Data Migration

Alternative Software Version

Permanent Service Restriction

Such changes may require governance through SCMS, SCFS, SDS, or other applicable SLGA standards.

Recovery urgency does not eliminate relevant lifecycle governance obligations.

Residual Risk

RRVM must identify material risks remaining after corrective execution.

Relevant elements include:

Residual Operational Degradation

Unverified Data State

Temporary Configuration

Dependency Exposure

Security Restriction

Monitoring Limitation

Reduced Redundancy

Known Recovery Limitation

Outstanding Corrective Action

Residual risk must not disappear from the governance record when recovery is declared complete or conditionally accepted.

Recovery Uncertainty

RRVM must preserve material uncertainty.

Examples include:

Unknown Data Integrity

Unverified Artifact Identity

Incomplete Infrastructure State

Uncertain Dependency Compatibility

Insufficient Monitoring Coverage

Unknown Physical-System Condition

Conflicting Recovery Evidence

Unknown conditions must remain distinguishable from verified recovery conditions.

Recovery Evidence Conflict

Recovery evidence may be materially inconsistent.

Examples include:

Rollback Pipeline Reports Success

Production Version Verification Reports Mismatch

Service Health Check Reports Normal

Data Integrity Verification Reports Failure

RRVM must preserve conflicting evidence and identify the affected recovery conclusion.

A favorable execution report must not override unresolved material evidence conflicts.

Recovery Reassessment

Recovery reassessment may be required when:

Corrective Execution Fails

Operational Conditions Change

New Evidence Emerges

Recovery Resources Become Unavailable

Dependencies Change

Authorization Changes

Additional Failures Appear

Monitoring Coverage Degrades

Residual Risk Increases

Recovery Objective Changes

Reassessment must identify whether the current strategy remains feasible, legitimate, and appropriate.

Recovery State Classification

RRVM recognizes the following governance classifications:

Recovery Not Initiated

Recovery Assessment Required

Under Assessment

Recovery Objective Defined

Strategy Selection Pending

Feasibility Not Established

Authority Not Established

Prerequisites Unresolved

Recovery Authorized

Recovery In Progress

Recovery Paused

Rollback Executed

Verification Pending

Partially Recovered

Recovered With Restrictions

Recovery Verified

Recovery Failed

Alternative Recovery Required

Evidence Insufficient

Recovery Conflicted

Reassessment Required

Closed With Residual Conditions

Closed

Unknown

Classification must remain associated with the relevant recovery case, environment, observation time, and supporting evidence.

Recovery Closure

Recovery closure requires an explicit governance conclusion.

Relevant elements include:

Recovery Objective

Corrective Actions Performed

Resulting Production Version

Resulting Operational State

Verification Results

Residual Conditions

Outstanding Actions

Applicable Restrictions

Closure Authority

Closure Evidence

Closure must not be inferred solely from the absence of additional alerts.

A recovery case may close with documented residual conditions where legitimate acceptance requirements permit it.

Historical Recovery Continuity

RRVM must preserve relevant recovery history.

This includes:

Original Deployment State

Failure Trigger

Initial Recovery Objective

Selected Strategy

Authorization Decisions

Corrective Attempts

Failed Recovery Actions

Alternative Strategies

Resulting Operational States

Verification Outcomes

Residual Conditions

Closure Decision

Historical continuity supports later operational reassessment,

accountability, and software lifecycle governance.

Governance Requirements

A conforming RRVM implementation must satisfy the following requirements. Identity and Assessment

- Establish a unique recovery case identity
- Associate the case with an identifiable deployment
- Preserve the corrective-action trigger
- Identify the affected operational environment
- Establish the current operational state
- Preserve relevant historical state references
- Define the recovery objective
- Identify material uncertainties

Strategy and Feasibility

- Identify applicable rollback and recovery strategies
- Evaluate strategy suitability
- Evaluate rollback feasibility
- Evaluate recovery feasibility
- Identify irreversible consequences
- Identify mandatory recovery prerequisites

• Evaluate data compatibility

• Evaluate configuration and dependency compatibility

• Identify service continuity requirements

• Identify recovery resources and capabilities

Authority and Execution

• Identify applicable corrective-action authority

• Verify decision actor eligibility

• Verify execution actor eligibility

• Define permitted corrective actions

• Preserve tool-path and constituent action boundaries

• Identify emergency authorization conditions

• Establish execution checkpoints

• Establish recovery stop conditions

• Preserve corrective execution evidence

• Prevent technical capability from substituting for legitimate authority.

Verification and Residual Conditions

• Establish recovery verification criteria

• Identify the resulting production version

• Evaluate resulting operational conditions

• Preserve monitoring evidence

• Identify partial recovery

• Identify restricted recovery states

• Preserve residual risks

• Preserve evidence conflicts

• Identify recovery failure

• Establish alternative recovery requirements where necessary

Continuity and Closure

• Identify material reassessment triggers

• Preserve recovery strategy revisions

• Preserve historical corrective actions

• Establish recovery-state classification

• Preserve applicable lifecycle change references

• Establish evidence-supported recovery closure

• Preserve unresolved conditions

• Provide resulting state references to PVCM and DSMM

Governance Logic

Deployment Failure or Unacceptable State → Recovery Trigger → Recovery

Case → Current State Assessment → Recovery Objective → Rollback and Recovery Options → Reversibility Assessment → Feasibility → Authority → Prerequisites → Strategy Selection → Corrective Execution → Production Version Verification → Operational State Verification → Residual Condition Assessment → Recovery Classification → Closure / Further Recovery

Where rollback is feasible:

Current State → Rollback Target → Compatibility Assessment → Authorized

Rollback → Execution → Resulting State Verification

Where rollback is infeasible or insufficient:

Current State → Recovery Objective → Alternative Strategy → Authority and

Feasibility → Execution → Operational Verification

Where recovery fails:

Recovery Attempt → Failure Evidence → Updated Operational State →

Reassessment → Alternative Recovery / Restriction / Escalation

Where recovery is partial:

Corrective Execution → Partial Restoration → Residual Conditions →

Restricted Acceptance / Additional Recovery

RRVM must not classify recovery as verified solely because a corrective tool or procedure reports success.

Governance Outputs

The principal output is:

Software Deployment Recovery Profile

Supporting outputs may include:

Recovery Case Identity

Recovery Trigger Record

Deployment Association Record

Current Operational State Reference

Previous Operational State Reference

Recovery Objective Record

Rollback Feasibility Assessment

Recovery Feasibility Assessment

Reversibility Assessment

Corrective Strategy Record

Recovery Authorization Reference

Execution Actor Eligibility Record

Recovery Prerequisite Register

Corrective Execution Record

Production Version Restoration Reference

Recovery Verification Record

Partial Recovery Record

Restricted Recovery State Record

Recovery Failure Record

Residual Risk Register

Recovery Reassessment Record

Recovery State Classification

Recovery Closure Record

Historical Recovery Register

Minimum Implementation Framework

A minimum RRVM implementation must establish eight governance layers. Layer 1 — Recovery Identity

Identify the recovery case, affected deployment, operational environment,

and corrective-action trigger. Layer 2 — Operational Reality

Establish the current operational state, relevant previous states, and the

intended recovery objective. Layer 3 — Strategy and Feasibility

Evaluate rollback, forward recovery, alternative restoration,

reversibility, and material operational constraints. Layer 4 — Authority

Identify legitimate decision and execution authority, including relevant

automation and autonomous agent boundaries. Layer 5 — Corrective Execution

Establish prerequisites, execution scope, sequence, checkpoints, stop

conditions, and evidence requirements. Layer 6 — Verification

Verify the resulting production version, configuration, dependencies, data

integrity, service conditions, and applicable operational requirements. Layer 7 — Residual Conditions and Continuity

Preserve partial recovery, remaining risks, failed attempts, reassessment

triggers, and historical corrective states. Layer 8 — Governed Output and Closure

Establish the Software Deployment Recovery Profile, classify the recovery

result, preserve closure evidence, and return resulting state references to PVCM and DSMM.

The implementation may be manual, software-supported, automated, or agent-assisted.

The governance requirements remain applicable regardless of the technology

used to perform corrective actions.

Cross-Module Boundary

DPM → RRVM

DPM establishes rollback and recovery planning references.

RRVM evaluates their current feasibility, applicability, authority,

execution, and resulting operational state.

DAM → RRVM

DAM establishes deployment authorization and applicable authority

references.

RRVM evaluates the authority required for corrective actions, which may differ from the original deployment authorization.

PVCM → RRVM

PVCM establishes previous and current production version identities.

RRVM uses those references to evaluate rollback targets and verify

resulting version states.

DSMM → RRVM

DSMM identifies unacceptable deployment conditions and intervention

triggers.

RRVM governs the corrective response and returns resulting operational

state references for monitoring and verification.

RRVM → PVCM and DSMM

RRVM provides corrective execution and recovery evidence.

PVCM verifies the resulting production version.

DSMM evaluates the resulting deployment state and applicable operational

acceptance conditions.

Cross-Architecture Boundary

RRVM consumes relevant governance references from:

SCMS — Software Change Management Standard

Change requests, impact assessment, coordination, and lifecycle re-entry

when corrective actions introduce material software changes.

SCFS — Software Configuration Standard

Configuration baselines, artifact identities, and configuration restoration

references.

SDS — Software Dependency Standard

Dependency compatibility and restoration requirements.

SQS — Software Quality Standard

Quality verification and defect references.

SSS — Software Security Standard

Security conditions and remediation constraints.

SLCS — Software Lifecycle Control Standard

Lifecycle transition, exception, and continuity governance.

GOA™ — Governance Architecture

Authority and delegation references.

AGA™ — Accountability Governance Architecture

Responsibility and accountability references.

ASGA™ — Autonomous Systems Governance Architecture

Autonomous execution, intervention, escalation, and human override

boundaries.

RRVM does not replace these mechanisms.

It governs their applicability to deployment-related rollback and recovery.

Practical Applications

Enterprise Software Deployment

Governing rollback and recovery when production updates cause operational failures.

Continuous Delivery

Ensuring automated rollback procedures remain authorized, feasible,

evidence-supported, and operationally verified.

Cloud-Native Infrastructure

Coordinating software, configuration, dependency, data, and infrastructure

restoration across distributed systems.

Autonomous AI Software Operations

Preventing agents from executing unrestricted corrective actions merely

because they detect a failure or possess technical access.

Critical Infrastructure

Preserving recovery authority, service continuity, operational

restrictions, and verification where deployment failures have consequential effects.

Robotics and Embedded Systems

Governing firmware recovery and operational restoration where software changes may affect physical system behavior.

Data-Intensive Applications

Distinguishing application rollback from database compatibility, data

integrity, and actual service recovery.

Use Case 1 — Software Rollback Succeeds but the Database Remains

Incompatible

A software provider deploys a new application version that modifies the database schema.

The deployment causes critical application failures.

An automated pipeline rolls back the application binary to the previous

version.

The pipeline reports rollback success.

However, the previous application version cannot operate with the modified

database schema.

RRVM evaluates:

Failure Trigger → Current Data State → Previous Application Version →

Rollback Feasibility → Corrective Execution → Data Compatibility → Operational Verification

The assessment identifies:

• The previous software artifact was restored

• The database remains in a materially changed state

• The restored application is incompatible with the current database

• The rollback procedure completed technically

• The required operational state was not recovered

The appropriate classification is:

Rollback Executed — Recovery Failed.

Software Artifact Restored ≠ Operational State Restored.

Use Case 2 — An Autonomous Agent Attempts an Unauthorized

Infrastructure Recovery

An autonomous deployment agent detects that a production service has become unavailable.

The agent has permission to monitor deployment conditions and restart

application processes.

It determines that replacing the production database server may restore service.

The agent possesses infrastructure credentials capable of performing the

replacement.

However, its delegated authority does not include database infrastructure replacement.

RRVM evaluates:

Recovery Trigger → Proposed Strategy → Agent Authority → Constituent

Actions → Operational Consequences → Authorization Applicability

The assessment identifies:

• The service failure is supported by evidence

• The proposed recovery strategy requires a material infrastructure

• The agent has technical access to the required tool

• The action exceeds its delegated recovery authority

• Escalation to an applicable decision authority is required

action.

The appropriate classification is:

Authority Not Established — Recovery Escalation Required.

Recovery Objective Authority ≠ Authority for Every Recovery Action.

Use Case 3 — A Robotic Firmware Rollback Cannot Restore the Previous

Operational State

A robotics operator deploys new firmware to an autonomous mobile robot

fleet.

Several units develop unexpected movement-control behavior.

The operator attempts to restore the previous firmware version.

However, the update modified persistent controller parameters that are not automatically reversed by firmware replacement.

RRVM evaluates:

Firmware Deployment → Observed Physical Behavior → Previous Firmware →

Persistent State Changes → Reversibility → Recovery Strategy → Operational Verification

The assessment identifies:

• The previous firmware image remains available

• Firmware replacement is technically possible

• Persistent controller parameters have changed

• Restoring the earlier firmware alone cannot establish the previous

• Additional parameter restoration and physical-system verification are

operating conditions. required.

The appropriate classification is:

Recovery Feasibility Not Fully Established — Additional Restoration

Required.

Previous Firmware Restored ≠ Previous Physical Operating State Restored.

Critical Distinctions

Rollback Executed ≠ Operational Recovery Verified.

Previous Software Version Restored ≠ Previous Operational State Restored.

Recovery Procedure Available ≠ Recovery Feasible.

Technical Intervention Capability ≠ Legitimate Intervention Authority.

Service Restarted ≠ Service Recovered.

Recovery Trigger Detected ≠ Rollback Automatically Authorized.

Rollback Success ≠ Recovery Success.

Unknown ≠ Satisfied.

Backup Exists ≠ Backup Usable.

Exception Recorded ≠ Exception Authorized.

Recovery Capability ≠ Recovery Authority.

Recovery Objective Authority ≠ Authority for Every Recovery Action.

Corrective Execution Completed ≠ Recovery Objective Achieved.

Previous Artifact Available ≠ Previous State Reproducible.

Partial Recovery ≠ Complete Recovery.

Service Availability Restored ≠ Data Integrity Verified.

Recovery Verified With Restrictions ≠ Unrestricted Operational Acceptance.

Recovery Closure ≠ Historical Consequences Eliminated.

Canonical Boundary

RRVM governs the identification, evaluation, authorization applicability,

execution conditions, verification, and continuity of rollback and recovery actions addressing software deployment failures and unacceptable deployment-related operational states.

It establishes corrective-action triggers, recovery objectives, strategy

suitability, feasibility, authority, execution boundaries, operational verification, residual conditions, and recovery-state classification.

RRVM does not independently authorize original deployments, redefine software release validity, replace production version governance, or treat corrective procedure completion as proof of operational recovery.

Its principal governance output is the:

Software Deployment Recovery Profile

This output preserves the evidence-supported relationship between the

deployment failure, selected corrective action, resulting operational state, and remaining recovery obligations.

Canonical Closing Statement

A rollback is not successful merely because software returns to an earlier version. Recovery is successful only when the required operational state is legitimately and verifiably established.

A previous application binary may be restored while the database remains incompatible.

A service may restart while critical dependencies remain unavailable.

An automated pipeline may report recovery success while production contains an unexpected artifact.

An autonomous agent may identify an effective corrective strategy without possessing authority to execute it.

A robotic firmware update may be reversed while persistent controller changes continue affecting physical behavior.

And a recovery process may restore essential functionality while leaving restrictions, unresolved risks, or incomplete verification.

RRVM establishes the governance structure required to distinguish these

outcomes.

It requires corrective actions to be justified by identifiable operational

conditions, supported by feasibility evidence, and performed within legitimate authority and execution boundaries.

It preserves the distinction between rollback and recovery, between

technical execution and operational verification, and between restored functionality and complete resolution of deployment consequences.

It recognizes that recovery may require forward correction, partial restoration, restricted operation, or alternative strategies when reversal is impossible or insufficient.

It ensures that failed recovery attempts, irreversible consequences, residual risks, and historical operational states remain traceable.

And it connects corrective execution back to production version governance,

deployment-state monitoring, and the wider software lifecycle.

The objective of recovery governance is not to return software to an earlier point in time. It is to establish an acceptable operational reality through legitimate, feasible, controlled, and verifiable corrective action.

RRVM transforms rollback and recovery from technical emergency procedures

into accountable governance mechanisms for restoring operational integrity, preserving evidence, and maintaining software lifecycle continuity.